

Work fluently with integers, decimals, fractions and percentages

Step 41

Use different number forms flexibly and switch between them with purpose.

★ Today I am learning to work fluently with integers, decimals, fractions and percentages.

1. Convert it!

Convert between the given number forms.

a) Write each decimal as a fraction in simplest form.

1) $0.25 =$ _____

2) $0.6 =$ _____

3) $0.75 =$ _____

4) $0.125 =$ _____

b) Write each fraction as a decimal.

1) $\frac{1}{2} =$ _____

2) $\frac{3}{4} =$ _____

3) $\frac{2}{5} =$ _____

4) $\frac{7}{8} =$ _____

c) Write each decimal as a percentage.

1) $0.18 =$ _____ %

2) $0.5 =$ _____ %

3) $2.35 =$ _____ %

4) $0.007 =$ _____ %

d) Write each percentage as a decimal.

1) $35\% =$ _____

2) $72.5\% =$ _____

3) $120\% =$ _____

4) $3.6\% =$ _____

2. Relate and Compare

 Fill in the missing value using $<$, $>$ or $=$.

a) 0.75 _____ $\frac{3}{4}$

b) 25% _____ 0.2

c) $\frac{2}{5}$ _____ 40%

d) 1.5 _____ 150%

e) -3 _____ -2.5

f) 0.08 _____ 8%

g) $\frac{3}{10}$ _____ 0.25

h) 75% _____ $\frac{7}{10}$

i) -1.2 _____ -120%

j) $1\frac{1}{4}$ _____ 1.2

k) $\frac{5}{8}$ _____ 62.5%

l) 0.3 _____ $\frac{1}{3}$

Remember:

$\frac{1}{4} = 0.25 = 25\%$

$\frac{1}{2} = 0.5 = 50\%$

$\frac{3}{4} = 0.75 = 75\%$

$1 = 1.00 = 100\%$

3. Real-World Connections

Solve the problems. Show your working.

 a) A pizza costs \$18. Sarah eats $\frac{1}{4}$ of it. What percentage of the pizza does she eat?


Answer: _____

b) A jacket is on sale for 30% off. The original price is \$64.40. What is the sale price?



Answer: \$ _____

 c) A beaker contains 0.75 L of water. $\frac{1}{3}$ of the water is used. How many litres are left?


Answer: _____ L

d) A chocolate bar has 24 pieces. Mark eats 40% of the pieces. How many pieces did he eat?



Answer: _____

4. Choose the Best Form

Write each answer in the most useful form for the situation.

 a) The temperature is -2.4°C . Write this as:

- 1) an integer (approximate) _____
- 2) a fraction _____
- 3) a percentage of 0°C _____

 b) $\frac{3}{5}$ of a tank is full. Write this as:

- 1) a decimal _____
- 2) a percentage _____

c) A recipe uses 0.6 kg of flour. Write this as:

- 1) a fraction _____
- 2) a percentage _____

d) A student scored 18 out of 25. Write this score as:

- 1) a decimal _____
- 2) a percentage _____
- 3) a fraction in simplest form _____

5. Problem Solving Challenge

Solve these multi-step problems. Show your working.

a) A shop gives a 15% discount on a \$82 item. Then 8% GST is added to the sale price. What is the final price? \$ _____

 b) A class of 24 students: $\frac{1}{3}$ prefer Maths, 25% prefer Science and the rest prefer English. How many students prefer English?

Answer: \$ _____ students

Working space

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|--|---|
| ☐ Confident with all number forms | ☐ Uses different forms flexibly |
| ☐ Needs support with fractions | ☐ Checks answers carefully |
| ☐ Needs support with decimals | ☐ Works neatly and systematically |
| ☐ Needs support with percentages | ☐ Tries hard and shows improvement |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____